

VISION DOCUMENT (PA3-2026)

1. Introduction

Assignee	Reviewer	Editor
Van	Chien	Nghia

The project outlines the end-to-end management framework, strategic scope, technical execution, and operational timeline for the design, development, and launch of **LoveYou—an AI-enhanced web application** engineered to redefine digital matchmaking for modern Vietnamese users seeking meaningful, long-term romantic relationships. While global online dating solutions are widely accessible, they frequently rely on generic matching frameworks that overlook the distinct social dynamics, relationship expectations, and cultural values unique to the Vietnamese demographic. **LoveYou** project bridges this gap by merging machine learning technology with localized sociocultural insights, delivering a personalized, secure, and intuitive digital space where individuals can discover real compatibility based on shared life goals, communication preferences, and core personal values.

Ultimately, this project describes the overall approach, schedule, and management strategy for developing **LoveYou**, an AI-enhanced dating web application designed to help Vietnamese users find meaningful romantic connections online. It serves as the roadmap for the team during development and is used to coordinate work, track progress, and control delivery. Furthermore, **LoveYoy** will be served as the foundation for positioning as a trusted leader in Viet Nam's evolving digital relationship landscape.

4. Product Overview

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4.1 Product Perspective

Core Differentiation

LoveYou is developed as a modern web application following a client-server architecture. Users access the platform through a web browser, while all business logic, AI recommendation services. Unlike traditional platforms like Tinder or Bumble, which rely on random, low-context swipe mechanics that force snap, superficial decisions, LoveYou optimizes the user experience through two core pillars: **Web Client**

- Provides responsive user interfaces for registration, profile management, matching, messaging, notifications, and account settings.
- Communicates with the backend through RESTful APIs.

Backend Server

- Handles authentication and authorization.
- Processes user profiles and matchmaking requests.

- Calculates AI compatibility scores.

Database

- Stores user accounts, profiles, preferences, conversations, matches, notifications, reports, and system configurations.
- Maintains secure and consistent application data.

AI Matching Service

- Analyzes user interests, biographies, dating preferences, age ranges, and locations.

Core Functional Modules

Field	Details
Module ID	MOD-01
Name	Authentication, Authorization, Onboarding & Preference Setup Module
Description	Handles secure user onboarding, including sign up, log in, log out, forgot/reset password, session management, and role-based access control. It features a first-login wizard where users upload photos, set gender/age preferences, and select initial interest tags.

Field	Details
Module ID	MOD-02
Name	User Profile Management Module
Description	Empowers users to customize their profile by editing personal info, uploading an avatar and additional photos (up to 5), managing interest tags, and changing passwords.

Field	Details
Module ID	MOD-03
Name	AI-Powered Smart Matching Subsystem
Description	Calculates the AI compatibility score, displays a ranked candidate list, shows matching reasons in Vietnamese, and refreshes suggestions to deliver optimized potential matches.

Field	Details
Module ID	MOD-04
Name	Swipe & Match System
Description	Retains the familiar, intuitive gesture control allowing users to swipe right (like) or left (skip), automatically detects mutual likes to trigger a match, shows match confirmations, and maintains a match history.

Field	Details
Module ID	MOD-05
Name	Advanced Search & Filtering Module
Description	Provides users with the ability to filter candidates dynamically by gender, age range, city, and interest tags, showing paginated results sorted by recency or AI score.

Field	Details
Module ID	MOD-06
Name	Real-time Messaging & Notification Center
Description	Supports instant text communication with messaging history, online/offline status, and a typing indicator, while also offering an AI ice-breaking game to initiate conversations smoothly. It pushes real-time match and message notifications with precise timestamps.

Field	Details
Module ID	MOD-07
Name	Privacy, Safety Controls & Admin Dashboard
Description	Protects users with robust safety measures, such as blocking/reporting users, hiding contact details until a match is confirmed, deactivating or permanently deleting accounts, and utilizing AI to analyze chats for red flags . On the administrative side, it provides a comprehensive console to view platform statistics, manage users (search, view, block, delete), and curate the interest tag catalogue.

4.2 Assumptions and Dependencies

To ensure that the LoveYou platform operates reliably, achieves accurate matchmaking, and maintains high performance, the following technical assumptions and external dependencies have been identified:

Aspect	Assumption	Dependency / Impact
1. Internet Access	Users are assumed to have a stable internet connection (Wi-Fi, 4G, or 5G) when accessing the web application.	Unstable or unavailable connectivity will severely disrupt real-time messaging, typing indicators, and immediate push notifications.
2. User Engagement & Data Input	Users are expected to complete their profiles accurately, selecting true interest tags and providing honest biographical data during onboarding.	The accuracy of the AI-powered matching engine and the resulting Compatibility Score (0–100) heavily rely on the quality of user-provided profile data.
3. AI Model Capabilities	The core AI engine can process multi-dimensional data quickly and	System responsiveness when displaying the ranked candidate list or refreshing

Aspect	Assumption	Dependency / Impact
	generate human-readable match explanations in Vietnamese without noticeable delay.	suggestions directly depends on the availability and processing speed of the underlying AI processing infrastructure.
4. Device & Browser Compatibility	Users will access the application through standard modern web browsers on both desktop computers and mobile devices.	While the web interface is inherently responsive, the visual rendering and execution of dynamic features (like the AI ice-breaking game) depend on browser compatibility with modern web standards.
5. Admin Moderation & Governance	Platform administrators will actively monitor the dashboard, handle user reports, and maintain the interest tag catalog.	Maintaining a safe and high-quality dating ecosystem is highly dependent on swift admin actions following the detection of AI chat red flags or malicious user behavior.
6. Cloud Storage Infrastructure	The underlying cloud infrastructure will provide stable storage capable of scaling as data demands grow.	Seamless media loading and data continuity are dependent on the cloud infrastructure securely managing up to 5 photos per user profile, entire chat histories, and real-time logs.